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OPTIMIZATION RESULTS

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Roll	Stop	Win Rate	Avg Moves	Avg Time	
1	4	75,0	% 61,7	3089,4	ms
1	6	75,0	% 59,9	2998,8	ms
3	6	70,0	% 58,5	2930,6	ms
4	6	70,0	% 65,6	3285,6	ms
2	4	65,0	% 64,1	3209,5	ms
1	5	60,0	% 62,1	3111,4	ms
2	6	60,0	% 60,5	3030,7	ms
4	4	60,0	% 66,5	3328,7	ms
3	5	55,0	% 58,7	2940,6	ms
5	4	55,0	% 61,6	3085,2	ms
2	5	50,0	% 57,6	2883,6	ms
3	4	50,0	% 63,1	3158,6	ms
5	6	50,0	% 60,3	3020,9	ms
5	5	45,0	% 59,6	2983,6	ms
4	5	35,0	% 60,5	3030,2	ms

--- Top 3 Configurations ---

1. rollDepth=1, stopDepth=4: 75,0% win rate
2. rollDepth=1, stopDepth=6: 75,0% win rate
3. rollDepth=3, stopDepth=6: 70,0% win rate

--- Best Balanced (performance/time) ---

rollDepth=1, stopDepth=4: 75,0% win rate, 3089,4 ms avg

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Expectiminimax Depth Optimization
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Games per configuration: 20

Roll depth range: 1-5

Stop depth range: 4-6

Seed: 42

Time budget: 50 ms per move (iterative deepening)

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[1/15] Testing rollDepth=1, stopDepth=4... Win rate: 75,0% (avg moves: 61,7, avg time: 3089,4 ms)
[2/15] Testing rollDepth=1, stopDepth=5... Win rate: 60,0% (avg moves: 62,1, avg time: 3111,4 ms)
[3/15] Testing rollDepth=1, stopDepth=6... Win rate: 75,0% (avg moves: 59,9, avg time: 2998,8 ms)
[4/15] Testing rollDepth=2, stopDepth=4... Win rate: 65,0% (avg moves: 64,1, avg time: 3209,5 ms)
[5/15] Testing rollDepth=2, stopDepth=5... Win rate: 50,0% (avg moves: 57,6, avg time: 2883,6 ms)
[6/15] Testing rollDepth=2, stopDepth=6... Win rate: 60,0% (avg moves: 60,5, avg time: 3030,7 ms)
[7/15] Testing rollDepth=3, stopDepth=4... Win rate: 50,0% (avg moves: 63,1, avg time: 3158,6 ms)
[8/15] Testing rollDepth=3, stopDepth=5... Win rate: 55,0% (avg moves: 58,7, avg time: 2940,6 ms)
[9/15] Testing rollDepth=3, stopDepth=6... Win rate: 70,0% (avg moves: 58,5, avg time: 2930,6 ms)
[10/15] Testing rollDepth=4, stopDepth=4... Win rate: 60,0% (avg moves: 66,5, avg time: 3328,7 ms)
[11/15] Testing rollDepth=4, stopDepth=5... Win rate: 35,0% (avg moves: 60,5, avg time: 3030,2 ms)
[12/15] Testing rollDepth=4, stopDepth=6... Win rate: 70,0% (avg moves: 65,6, avg time: 3285,6 ms)
[13/15] Testing rollDepth=5, stopDepth=4... Win rate: 55,0% (avg moves: 61,6, avg time: 3085,2 ms)
[14/15] Testing rollDepth=5, stopDepth=5... Win rate: 45,0% (avg moves: 59,6, avg time: 2983,6 ms)
[15/15] Testing rollDepth=5, stopDepth=6... Win rate: 50,0% (avg moves: 60,3, avg time: 3020,9 ms)
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